MOHAMMAD IMRAN HOSSAIN

Marie Skłodowska-Curie Actions (MSCA) Doctoral Fellow: Medical AI and Computational Oncology

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RESEARCH PROFILE

MSCA Doctoral Researcher at Karolinska Institutet working at the intersection of medical imaging and artificial intelligence to address clinical challenges in personalized breast cancer screening. My research experience spans deep learning, medical image analysis, computational pathology, multimodal analysis, and precision medicine. I am particularly interested in methodological innovation, rigorous model validation, and translating trustworthy medical AI into clinically meaningful applications.

EDUCATION

PhD, Medical AI and Computational Oncology | Oct 2026 – present
Karolinska Institutet, Stockholm, Sweden; HER-CARE MSCA Doctoral Network (DC09)

MSc, Bioinformatics and Modeling | Sep 2024 – Sep 2025
Sorbonne Université, Paris, France; Grade: 14.85/20 (Ranked 1st out of 8; Best Master's Thesis Award)

Erasmus Mundus Joint MSc, Medical Imaging and Applications (MaIA) | Sep 2022 – Aug 2024
Université Bourgogne Europe (France), University of Cassino and Southern Lazio (Italy), University of Girona (Spain); Grade: 8.30/10

BSc, Electrical and Electronic Engineering | Feb 2018 – Jun 2022
United International University, Dhaka, Bangladesh; Grade: 3.97/4.00 (Ranked 1st out of 120; Summa Cum Laude Award)

RESEARCH EXPERIENCE

MSCA Doctoral Research Fellow – Medical AI and Computational Oncology | Oct 2026 – present
Karolinska Institutet, Stockholm, Sweden
Research Focus: AI-based breast cancer risk prediction and personalized screening.

- Developing and validating mammography-based AI risk prediction models using single-time-point and longitudinal imaging data, clinical risk factors, and epidemiological data to identify women at elevated short-term breast cancer risk who may benefit from supplemental screening.
- Evaluating AI-based risk stratification to identify women at low risk of developing breast cancer within five years and and investigating opportunities to extend screening intervals while assessing interval cancers and cancer subtype-specific outcomes.
- Investigating subsequent breast cancer risk following a negative supplemental breast MRI to identify low-risk subgroups who may benefit from personalized MRI follow-up intervals.
- Assessing model discrimination, calibration, clinical utility, and generalizability across populations using longitudinal breast imaging and cancer outcome data from the KARMA (Sweden) and EMBED (USA) cohorts.

Research Engineer – Bioinformatics and Artificial Intelligence | Oct 2025 – Sep 2026
Institut Curie & PSL Research University, Paris, France

- Developed computational pipelines for analyzing spatial transcriptomics (10x Genomics Visium HD) and histopathology data to characterize the tumor microenvironment in breast (n = 8) and lung (n = 6) cancer cohorts from Institut Curie Hospital.
- Implemented reproducible workflows integrating unsupervised clustering at bin and cell levels (graph-based and deep learning methods), cell-type deconvolution, and spatial mapping to investigate cellular heterogeneity, spatial organization, cell–cell interactions, and immune-related molecular programs.
- Collaborated with multidisciplinary research teams to translate computational findings into biological insights and conducted bioinformatics workshops to support knowledge sharing and research activities.

Research Intern – Spatial Transcriptomics | Feb 2025 – Aug 2025
Institut Curie, Paris, France (in collaboration with Sanofi) | Master's Thesis Project
Research Project: Identification of Tumor Gene Signatures Underlying Fibroblast-Mediated T Cell Exclusion in Human Lung Cancer Using Spatial Transcriptomics.

- Developed end-to-end computational workflows for spatial transcriptomics data (10x Genomics Visium V2 and Visium HD) analysis, including quality control, graph-based unsupervised clustering (Leiden), cell-type deconvolution (cell2location), and annotation using statistical and deep learning approaches across two independent cohorts (n = 6 samples per platform).
- Identified eight tumor-associated candidate genes potentially mediating cancer-associated fibroblast (CAF)-driven immunosuppressive barriers in lung squamous cell carcinoma; findings are being used for in vitro validation and contributing to an ongoing manuscript in preparation.

Research Intern – Computational Pathology | Feb 2024 – Jul 2024
French National Centre for Scientific Research (CNRS), Dijon, France | Master's Thesis Project
Research Project: Prediction of Homologous Recombination Deficiency (HRD) from Breast and Ovarian Cancer Whole-Slide Histopathology Images Using Deep Learning.

- Implemented and benchmarked deep learning approaches for HRD biomarker prediction from whole-slide H&E images across the TCGA and independent cohorts: fully supervised CNNs, weakly supervised attention-based multiple instance learning methods (AB-MIL, CLAM, HIPT, DS-MIL, Trans-MIL), and self-supervised transfer learning using foundation models (Phikon).

- Achieved the best performance among the evaluated models using TransMIL with Phikon (ROC-AUC: 0.78 for breast cancer, 0.68 for ovarian cancer); a systematic comparison of MIL architectures highlighted limitations of unimodal approaches and motivated future multimodal integration strategies.

Visiting Researcher – Medical Image Analysis | Aug 2023 – Oct 2023
Radboud University Medical Center, Nijmegen, Netherlands
Research Project: Real-Time Magnetic Resonance Image Reconstruction for Interventional Radiology Using Deep Learning.

- Developed a preprocessing pipeline to extract and undersample k-space data from raw MRI scans for real-time MRI reconstruction.
- Evaluated deep learning (U-Net) models on 25% undersampled k-space data to enhance reconstruction accuracy for interventional radiology.

SELECTED MACHINE LEARNING AND MEDICAL IMAGING PROJECTS

Automatic Brain Tissue Segmentation from MRI | University of Girona | MAIA Segmentation Challenge | 2023

- Developed pipelines for gray matter, white matter, and cerebrospinal fluid segmentation on IBSR18 brain MRI, incorporating N4 bias-field correction, anisotropic diffusion, normalization, patch extraction, and data augmentation. Compared probabilistic atlas-based segmentation with U-Net, LinkNet, and 2D/3D nnU-Net. The 3D nnU-Net achieved a mean Dice score of 0.937 ± 0.012; the project placed second in the MAIA brain tissue segmentation challenge. [github code]

Cardiac Structure Segmentation from 2D Echocardiograms | Sorbonne Université | 2025

- Developed and compared U-Net and nnU-Net pipelines for segmentation of the left ventricular endocardium, left ventricular epicardium, and left atrium in 2D echocardiography. nnU-Net achieved Dice scores of 0.94, 0.91, and 0.93, respectively, outperforming U-Net. Reported boundary errors were approximately 8–9 mm for HD95 and 0.28–0.44 mm for mean absolute distance.

Skin Lesion Classification with Machine Learning and Deep Learning | University of Girona | MAIA CADx Challenges | 2023

- Developed binary and multiclass classification pipelines using handcrafted features, feature selection, and classifier ensembles; implemented deep learning approaches based on transfer learning with ResNet/ResNeXt, DenseNet, EfficientNet, and Swin transformers. Used normalization, geometric and color augmentation, and prediction-averaging ensembles. The conventional machine learning pipeline achieved binary test accuracy of 0.8590 and multiclass test Cohen's kappa of 0.74. The best deep learning ensemble achieved a validation accuracy of 0.9070 for binary classification and a Cohen's kappa of 0.9448 for multiclass classification. These validation results are distinct from the conventional pipeline's test results. Both approaches placed second in their respective MAIA CADx challenges. [github code]

Deformable Lung CT Registration | University of Girona | 2023

- Developed pipelines to align inspiratory and expiratory lung CT scans from four COPDgene cases, including intensity preprocessing and lung segmentation. Compared affine and B-spline registration with Elastix/Transformix against VoxelMorph. Mean target registration error was 2.08 ± 2.26 mm for the intensity-based approach and 2.18 ± 2.74 mm for VoxelMorph; limited training data constrained the learning-based approach. [github code]

Breast Cancer Molecular Subtype Classification from Multi-omics Data | Sorbonne Université | 2024

- Integrated DNA methylation, copy-number variation, mRNA, and miRNA data from TCGA-BRCA for molecular subtype classification. Implemented exploratory data analysis, preprocessing, early data fusion, and feature selection, and compared six classifiers. XGBoost achieved the best balanced multiclass accuracy of 0.82.

Mammographic Breast Mass Detection | University of Cassino and Southern Lazio | 2023

- Developed an end-to-end mammographic mass detection pipeline combining multi-scale morphological shifting, mean shift filtering, k-means segmentation, feature extraction, and machine learning classification using the uCAS ML framework. Reported sensitivity, specificity, and accuracy were 0.96, 0.95, and 0.95, respectively. Presented the work as a poster at the 17th EXCITE Summer School on Biomedical Imaging.

RESEARCH MANUSCRIPTS

- Hossain, M. I., Amin, M. Z., Anyimadu, D. T., & Suleiman, T. A. (2024). Comparative Study of Probabilistic Atlas and Deep Learning Approaches for Automatic Brain Tissue Segmentation from MRI Using N4 Bias Field Correction and Anisotropic Diffusion Pre-processing Techniques. *ArXiv*. https://arxiv.org/abs/2411.05456 (Preprint).
- Tweneboah, A. D., Ahmed, S. T., & Imran, H. M. (2025). Deep learning and classical computer vision techniques in medical image analysis: Case studies on brain MRI tissue segmentation, lung CT COPD registration, and skin lesion classification. *ArXiv*. https://arxiv.org/abs/2502.19258 (Preprint).

SPECIALIST TRAINING AND RESEARCH PRESENTATIONS

17th EXCITE Summer School on Biomedical Imaging | ETH Zürich & University of Zürich, Switzerland | 2023

- Completed specialist training on deep learning and biomedical imaging and obtained 4 ECTS credits. Presented a research poster on mammographic breast mass detection.

FELLOWSHIPS, SCHOLARSHIPS, AND ACADEMIC HONORS

Marie Skłodowska-Curie Actions (MSCA) Doctoral Fellowship | European Union | 2026 – 2030

- Awarded a fully funded doctoral fellowship under the EU Horizon Europe MSCA HER-CARE Doctoral Network. Funding includes employment salary, mobility allowance, and family allowance in accordance with MSCA funding regulations and Swedish employment conditions.

Erasmus Mundus Scholarship | European Union | 2022 – 2024

- Awarded a fully funded scholarship for the Erasmus Mundus joint master's programme. The scholarship covered tuition fees, monthly living stipends, travel allowance, and installation costs. (total value of approximately €42,000)

Best Master's Thesis Award | Sorbonne Université | 2025

- Recognized for outstanding master's thesis research in bioinformatics and computational oncology on spatial transcriptomics and fibroblast-mediated T-cell exclusion in lung cancer with the highest distinction in French university grading systems.

Second Place — MAIA Research Challenges | University of Girona | 2023

- Placed second in brain tissue segmentation from MRI and skin lesion classification challenges.

Stipendium Hungaricum Scholarship | Hungarian Government | 2022 – 2024

- Awarded a fully funded scholarship for master's studies at Budapest University of Technology and Economics (BME). The scholarship includes tuition coverage and a monthly stipend.

Undergraduate Merit Scholarship & Summa Cum Laude | United International University | 2018 – 2022

- Received a 100% tuition fee waiver for 12 consecutive trimesters, covering the entire undergraduate degree. Received Summa Cum Laude distinction due to outstanding academic results (top 1%).

TEACHING AND RESEARCH MENTORING

Master's Intern Supervision | Institut Curie, Paris, France | 2026

- Co-supervised a master's thesis student, providing research guidance, technical support, and assistance with research activities, including reviewing code and documentation.

Undergraduate Teaching Assistant | United International University, Dhaka, Bangladesh | 2020 – 2022

- Supported more than 100 students through laboratory instruction, technical demonstrations, and practical coursework assistance. Helped students apply theoretical concepts during hands-on laboratory activities.

PROFESSIONAL AFFILIATIONS, LEADERSHIP, AND VOLUNTEERING

Member – Marie Curie Alumni Association (MCAA)
Program Representative – Erasmus Mundus Association (EMA)
Advisor & Former Vice President – Students' International Affairs Society (SIAS), United International University
Former General Member – IEEE, IEEE Bangladesh, and IEEE UIU Student Branch

TECHNICAL SKILLS

Machine Learning Frameworks: PyTorch; TensorFlow/Keras; Hugging Face; scikit-learn.
Representation Learning and Modeling: CNNs; U-Net and nnU-Net; vision transformers; vision foundation models; transfer learning; self-supervised representation learning (including DINO, masked image modeling, and iBOT); multiple-instance learning (including AB-MIL and TransMIL); multimodal modeling; diffusion models.
Medical Imaging and Computer Vision: MONAI; OpenCV; OpenSlide; ITK-SNAP; Elastix. Research applications include whole-slide image analysis, segmentation, deformable registration, MRI reconstruction, and image classification.
Computational Biology and Spatial Analysis: Seurat; Scanpy; Squidpy; SpatialData. Experience with 10x Visium V2 and Visium HD, scRNA-seq, quality control, cell-type deconvolution, spatial clustering, cell-cell communication, trajectory analysis, tumor microenvironment analysis, and integration of histopathology, bulk/scRNA-seq, and spatial transcriptomics.
Programming and Scientific Computing: Python; R; MATLAB; Bash; C/C++; CUDA. Linux and high-performance computing; GPU computing; Docker; Singularity; Git, GitHub, and GitLab.